

I2S 3-watt Class D amplifier, 3-watt speaker, and a few other components.

First, the credentials are configured, and then the I2S pins are defined for the amplifier. Fig. 3 shows a snippet of setting credentials.

```
ESP32_google_TTS_short.ino
1  #include "WiFi.h"
2  #include "HTTPClient.h"
3  #include "Audio.h"
4
5  // WiFi credentials
6  const char* ssid = "beral";
7  const char* password = "xxxxxxxxxx";
8
9  // I2S Pin Definitions
10 #define I2S_DOUT 25 // Data Out Pin
11 #define I2S_BCLK 33 //27 // Bit Clock Pin
12 #define I2S_LRC 32 //26 // Left Right Clock Pin
13
14 Audio audio;
15
16 void setup() {
17   Serial.begin(115200);
18
19   // Connect to WiFi
20   WiFi.begin(ssid, password);
21   while (WiFi.status() != WL_CONNECTED) {
22     delay(1000);
23     Serial.println("Connecting to WiFi...");
24   }
25   Serial.println("Connected to WiFi");
```

Fig. 3: Setting credentials

```
void playLongText(String text) {
  int maxChunkSize = 200; // can try with 265, Google TTS max length 200
  int startPos = 0;

  while (startPos < text.length()) {
    int endPos = min(startPos + maxChunkSize, (int)text.length()); // Cast text.length() to int
    String chunk = text.substring(startPos, endPos);

    // URL encode the text chunk
    chunk.replace(" ", "%20");

    // Construct the URL for Google TTS
    String tts_url = "http://translate.google.com/translate_tts?ie=UTF-8&q=" + chunk + "&tl=en&client=tw-ob";
    // Play the audio
    audio.connecttohost(tts_url.c_str());

    // Wait for the chunk to finish playing
    while (audio.isRunning()) {
      audio.loop();
    }
  }
}
```

Fig. 4: Setting the Google TTS URL

The code connects to the internet using Wi-Fi authentication. Then, it sends the input string to a Google TTS site to process it into speech. The longest sentence that comprises 265 characters can be spoken in one go, but up to 200 can be spoken for sure.

The same is shown in the first small sketch. The crux of the code is shown in Fig. 4.

Construction and testing

First, upload the source code into the ESP32 board. Then, connect the amplifier to the I2S pins on the ESP32 as per the circuit diagram. Connect the speaker to the output of the amplifier. For power, USB can be used, as input will need to be provided from

the serial monitor to the board to convert the sentence into speech. Therefore, power the ESP32 board by connecting it to a USB port.

Now, enter a sentence up to 200 characters on the serial monitor, which can be heard on the speaker. This does not mean that it will successfully speak a 200-character-long speech continuously, one after another. It may miss some parts due to the limited memory capacity of the ESP32. However, the first text always plays correctly.

For serial input-based speech to text, a similar modified sketch is also created within this device. It takes a string from the serial console and converts it into speech. However, small sentences comprising 3 to 4 words can be spoken continuously. Different combinations can be experimented with in the second sketch.

A working demo for this device can be found in the following link: <https://www.youtube.com/watch?v=18DOqX1lUGI> **EFY**



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EFY Note
The source code of this project is available for free download at www.electronicsforu.com